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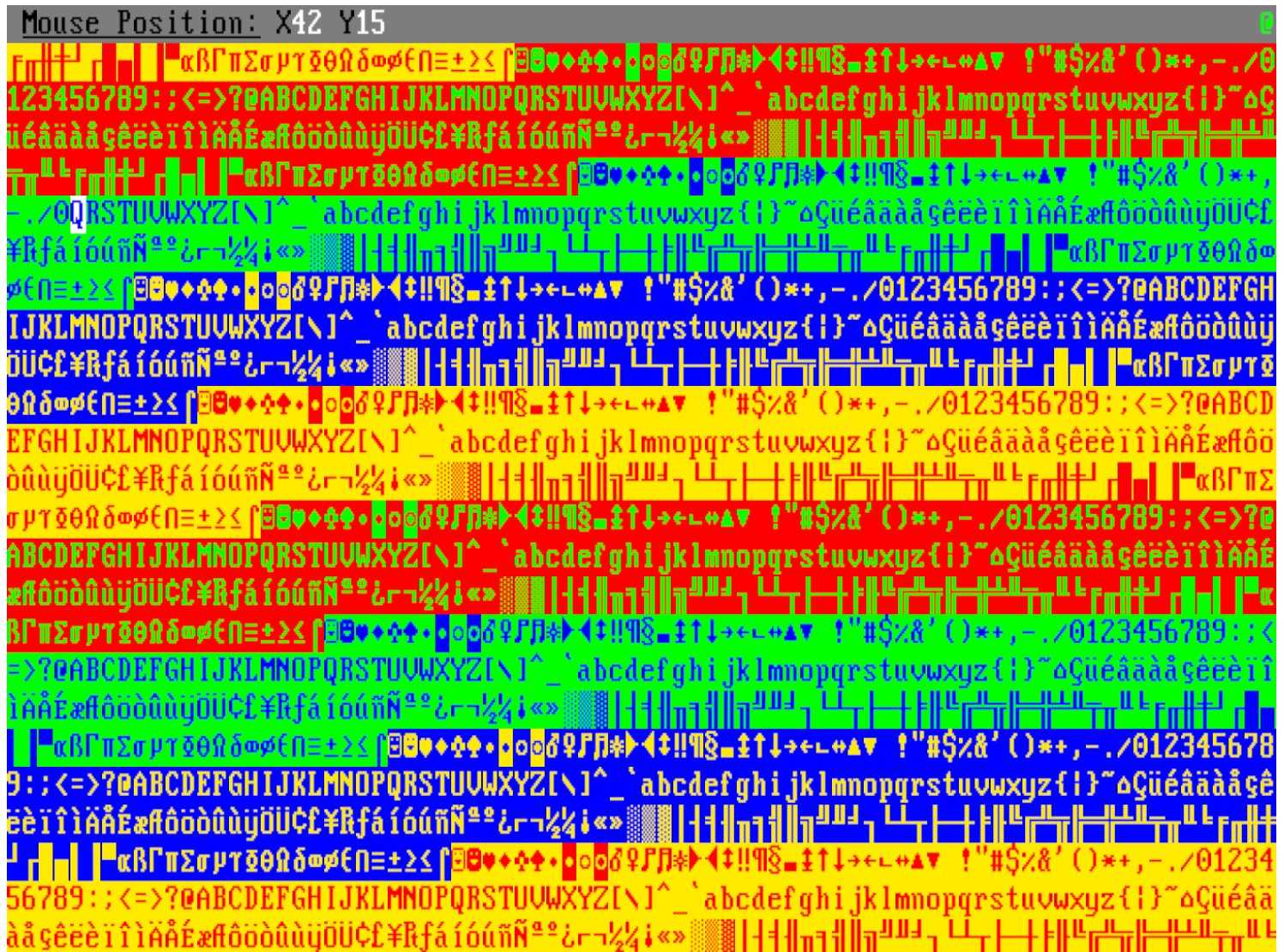
Sample Script Documentation

Introduction

This document is a brief overview of script SampleScript.cs provided in the package. Detailed setup information can be found in the Quick Start guide. More detailed framework information can be found in the API Documentation file.

Overview

The sample scripts shows the of usage of the majority of UCurses functionality. These include: Printing characters, printing strings, printing integers, enabling underline and blink, and retrieving the position of the mouse.



Screenshot of Sample Script

Code Breakdown

```
private int _currentColor = 0;
private int _currentAscii = 1;
private Vector2Int _currentPos = new Vector2Int(0,1);
private UnityEngine.Color[] _colorSet = new UnityEngine.Color[] { Color.red, Color.green, Color.blue, Color.yellow };
```

Variables used in the sample script. Contains the variables for keeping track of character and background colors, current position of the cursor, and an array of of basic colors.

```
void Start()
{
    UCurses.Instance.printBackgroundRectangle(new Vector2Int(0, 0), Color.gray, UCurses.Instance.GridSize.x, 1);
    UCurses.Instance.printString(new Vector2Int(1, 0), "Mouse Position: X", Color.black);
    UCurses.Instance.characterUnderlined(new Vector2Int(1, 0), true, 15);
    UCurses.Instance.printCharacter(new Vector2Int(21, 0), 'Y', Color.black);
    UCurses.Instance.printCharacter(new Vector2Int(UCurses.Instance.GridSize.x - 1, 0), '@', Color.green);
    UCurses.Instance.characterBlink(new Vector2Int(UCurses.Instance.GridSize.x - 1, 0), true, 0.5f);
    InvokeRepeating(nameof(drawTest), 0.01f, 0.01f);
}
```

Prints up all elements that don't change during runtime: Top gray border, underlined text at top of screen, and blinking character at top of screen.

Starts the a repeating drawTest() method.

```
private void Update()
{
    UCurses.Instance.printNumberInt(new Vector2Int(18, 0), UCurses.Instance.mousePosition().x, Color.white, 2, false);
    UCurses.Instance.printNumberInt(new Vector2Int(22, 0), UCurses.Instance.mousePosition().y, Color.white, 2, false);
}
```

Prints Int numbers at the top of the screen for the X and Y mouse position on the grid.

```
private void drawTest()
{
    UCurses.Instance.printBackground(_currentPos, _colorSet[_currentColor]);
    UCurses.Instance.printBackground(nextGridSpot(_currentPos), Color.white);
    UCurses.Instance.printCharacter(_currentPos, UCurses.Instance.asciiToChar(_currentAscii), _colorSet[nextColor(_currentColor)]);
    _currentPos = nextGridSpot(_currentPos);
    _currentAscii = nextAscii(_currentAscii);
}
```

Increments a white background cursor on the screen, moving top to bottom and left to right. Each time the cursor moves print a character in sequence. Once all 245 characters are printed change the color of the characters and background, and restart the sequence.

```
private int nextColor(int colorValue)
{
    colorValue++;
    if (colorValue == _colorSet.Length)
    {
        colorValue = 0;
    }
    return colorValue;
}
```

Returns the next color value in sequence, resets once all colors are used.

```
private int nextAscii(int asciiValue)
{
    asciiValue++;
    if (asciiValue == 245)
    {
        asciiValue = 1;
        _currentColor = nextColor(_currentColor);
    }
    return asciiValue;
}
```

Returns the next character ASCII value in sequence, resets once all values are used.

```
private Vector2Int nextGridSpot(Vector2Int current)
{
    current.x++;
    if (current.x == UCurses.Instance.GridSize.x)
    {
        current.x = 0;
        current.y++;
        if (current.y == UCurses.Instance.GridSize.y)
        {
            current.x = 0;
            current.y = 1;
        }
    }
    return current;
}
```

Finds the next position for the cursor. Moves left until it hits the edge of the screen, then moves down one space. Wraps around to the starting position once the cursor reaches the bottom right.